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RESEARCH ARTICLE

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Meta-analysis: Risk of hyperhidrosis with second-generation antidepressants

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Background: Our goal was to quantify the risk of hyperhidrosis associated with commonly used antidepressant agents and examine the impact of medication class, pharmacodynamics, and dose on risk of hyperhidrosis.

Methods: We conducted a PubMed search to identify all double-blind, randomized, placebo-controlled trials examining the efficacy of second-generation antidepressant medications in the treatment of adults with a depressive disorder, anxiety disorders, or obsessive-compulsive disorder. We used a random-effects meta-analysis to examine the pooled risk ratio of hyperhidrosis reported as a side effect in adults treated with second-generation antidepressants compared to placebo. We used stratified subgroup analysis and metaregression to examine the effects of medication type, class, dosage, indication, and receptor affinity profile on the measured risk of hyperhidrosis.

Results: We identified 76 trials involving 28,544 subjects. There was no significant difference in the risk of hyperhidrosis between serotonin–norepinephrine reuptake inhibitors (SNRI) [risk ratio (RR) = 3.17, 95% CI: 2.63–3.82] and selective serotonin reuptake inhibitors (SSRI) (RR = 2.93, 95% CI: 2.46–3.47) medications compared to placebo. All antidepressant medications were associated with a significantly increased risk of hyperhidrosis except fluvoxamine (RR = 0.56, 95% CI: 0.12–2.53), bupropion (RR = 1.23, 95% CI: 0.57–2.67), and vortioxetine (RR = 1.35, 95% CI: 0.79–2.33). The dose of SSRI/SNRI medications was not significantly associated with the risk of hyperhidrosis. Increased risk of hyperhidrosis was associated with increased affinity of SSRI/SNRI medications to the dopamine transporter.

Conclusion: Risk of hyperhidrosis was significantly increased with most antidepressant medications but was associated with dopamine transporter affinity.

KEYWORDS

hyperhidrosis, meta-analysis, second-generation antidepressants, serotonin uptake inhibitors, sweating

1 | INTRODUCTION

Selective serotonin reuptake inhibitors (SSRIs) and serotonin–norepinephrine reuptake inhibitors (SNRIs) are first-line pharmacological treatments for depressive disorders, obsessive-compulsive disorder (OCD), and anxiety disorders (Bose, Korotzer, Gommoll, & Li, 2008; Coric et al., 2010; Lepola, Loft, & Reines, 2003; Mendels, Kiev, & Fabre, 1999; Montgomery, Kasper, Stein, Hedegaard, & Lemming, 2001; Stein, Wreford Andersen, Tonnoir, & Fineberg, 2007; Suliman et al., 2015). Atypical antidepressant agents, such as bupropion, mirtazapine, vortioxetine, and vilazodone are also utilized as potential first-line treatments for depressive disorders (Baldwin, Ajel, & Garner, 2010; Dell'Osso, Nestadt, Allen, & Hollander, 2006; National Guideline, 2010; Reinhold, Mandos, Rickels, & Lohoff, 2011; Thase, 2006; Weilburg, 2004; Whooley & Simon, 2000). Although these medications are generally well tolerated, adverse effects are common and are often the primary cause of medication discontinuation or dose reduction (National Guideline, 2010). Roughly one-third to one-half of patients discontinue pharmacological treatment (Lader, Stender, Bürger, & Nil, 2004; S. Kasper, 2005) with a quarter of those patients reported

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side effects to be the reason for their discontinuation (Lader et al., 2004). Meta-analysis has demonstrated that higher doses of SSRIs are modestly more effective than lower doses of SSRIs for depressive and anxiety disorders as well as OCD (Issari, Jakubovski, Bartley, Pittenger, & Bloch, 2016; Jakubovski, Varigonda, Freemantle, Taylor, & Bloch, 2016). On the other hand, higher doses of antidepressants are associated with higher rate of dropout due to side effects (Issari et al., 2016; Jakubovski et al., 2016). Although these higher doses are associated with higher rates of dropout, many side effects of antidepressants are likely not dose dependent. Therefore, reducing dosing of antidepressant medications often does not eliminate or even reduce side effects. Better understanding of likelihood of side effects of common antidepressant medications and their relationship to individual agents and antidepressant dosing will help clinicians to make more evidence-based treatment decisions when side effects occur.

Antidepressant-induced excessive sweating (ADIES), or hyperhidrosis, occurs in 5–14% of patients taking antidepressants overall (Mago, Thase, & Rovner, 2013), approximately 10% of patients treated with SSRIs, and 5–20% of patients treated with SNRIs (Baldwin et al., 2010; Dell'Osso et al., 2006). Hyperhidrosis can be embarrassing and result in changes in daily activity. Furthermore, excessive sweating can often worsen underlying symptoms of an existing anxiety disorder, and this combination can contribute to patients discontinuing their antidepressant medications. Although limited data exist, roughly 2% of patients prescribed antidepressants may discontinue the medication due to hyperhidrosis (Marcy & Britton, 2005). Furthermore, hyperhidrosis often appears to be a long-term side effect of antidepressant medication that does not remit with continued treatment (Mavissakalian, Perel, & Guo, 2002).

Sweating or perspiration is primarily thermoregulatory, and while the exact mechanism by which antidepressant medications induce hyperhidrosis is unclear, it is clear that it involves a degree of central dysregulation of this thermoregulatory mechanism. Several neurotransmitters affected by antidepressants are important in thermoregulation. Serotonin may exert a direct effect on hypothalamic serotonin receptors involved in thermoregulation (Marcy & Britton, 2005). The relationship between serotonin and thermoregulation is supported by the observation that serotonin syndrome is associated with an increased core temperature and increased sweating. Impulses from the hypothalamus travel through an autonomic pathway to the spinal cord and then to the skin via the sympathetic nervous system. Sweat glands are primarily innervated by cholinergic nerve fibers and therefore antidepressant effects on adrenergic neurotransmitters (e.g., norepinephrine, acetylcholine, and dopamine) may also have direct effects on sweating in the peripheral nervous system (Marcy & Britton, 2005).

Understanding differences in the risk of hyperhidrosis between different antidepressant agents may (1) aid in understanding the primary underlying mechanism for this side effect; (2) aid in the selection of effective treatments for ADIES or hyperhidrosis, and (3) help guide optimal management of this side effect and underlying psychiatric illness when both occur. Previous meta-analyses have compared the risk of hyperhidrosis among commonly prescribed antidepressant medications but have relied exclusively on the incidence of hyperhidrosis within the active medication group (and not normalizing data to the reported incidence of hyperhidrosis in the placebo group). This methodology is extremely vulnerable to confusing the different methods of reporting side effects between trials across agents with actual differences between agents. Nonetheless, these previous meta-analyses have suggested that tricyclic antidepressants are most likely to cause hyperhidrosis, while the differences in risk between SRI agents are unclear (Baldwin et al., 2010; Trindade, Menon, Topfer, & Coloma, 1998).

The goals of the current meta-analysis are to determine (1) the magnitude of risk of hyperhidrosis associated with individual pharmacological agents; (2) if there is a difference in the risk of hyperhidrosis between antidepressant classes; (3) if there is a difference in the risk of hyperhidrosis within antidepressant classes; (4) if there is a relationship between antidepressant dose and the risk of hyperhidrosis; (5) if there is a relationship between diagnostic indication for the use of second-generation antidepressants and the risk of hyperhidrosis; (6) whether pharmacological properties of an antidepressant medications (e.g., receptor affinities) are associated with the risk of hyperhidrosis.

2 | METHODS

2.1 | Search strategy for identification of studies

Two reviewers searched the electronic database of PubMed on June 10, 2016 for relevant studies using the following search: (fluoxetine OR fluvoxamine OR citalopram OR escitalopram OR sertraline OR paroxetine OR bupropion OR trazodone OR desvenlafaxine OR duloxetine OR mirtazapine OR venlafaxine OR vilazadone OR vortioxetine) OR (serotonin uptake inhibitors) AND (Depressive disorder OR anxiety disorders) AND (Humans). The search was limited to randomized controlled trials. The references of relevant systematic reviews and meta-analyses were searched for relevant published and unpublished research.

2.2 | Selection of studies

The titles and abstracts of studies obtained by this search strategy were examined by two reviewers (C.B. and K.C.) to determine inclusion in this meta-analysis. Any discrepancies were resolved by a final reviewer (M.H.B.). Inclusion criteria for this study were: (1) randomized, double-blind, (2) placebo-controlled clinical trials of (3) a second-generation antidepressant of interest (fluoxetine, fluvoxamine, citalopram, escitalopram, sertraline, paroxetine, bupropion, trazodone, desvenlafaxine, duloxetine, mirtazapine, venlafaxine, vilazadone, and vortioxetine); targeting (4) clinical symptoms of major depressive disorder, anxiety disorder, or obsessive compulsive disorder. Exclusion criteria for this study were: (1) discontinuation studies; (2) head-to-head studies between antidepressants without a placebo control; (3) crossover trials; (4) studies requiring concomitant medications; (5) non treatment trials.

2.3 | Meta-analytic procedures

Data were extracted by independent reviewers (C.B. and K.C.) on specifically developed spreadsheets on Microsoft Excel. Our primary outcome measure was the proportion of patients or subjects in a trial (n) reporting hyperhidrosis as a side effect of medication, compared to the total number of patients or subjects in a trial (N). Data were also gathered on trial medication, trial design, maximum daily medication dose, diagnostic indication for second-generation antidepressant, and other relevant attributes and results of the trials. Any disagreement among reviewers was mitigated through discussion and the procurement of more information from the study investigators if possible. When agreement could not be attained between the initial reviewers, the senior investigator (M.H.B) resolved all disputes.

All statistical analyses were completed in Comprehensive Meta-analysis Version 3 (Biostat, 2015). For our outcome measures of interest, proportion of participants experiencing hyperhidrosis was analyzed using pooled RR. For all outcome measures, 95% CIs were conveyed. A random-effects model for meta-analysis was used. Publication bias was assessed by plotting the effect size against standard error for each included trial (i.e., funnel plot). In addition, publication bias was statistically tested by the Egger test. When publication bias was present, we also present sensitivity analysis where we adjusted for publication bias using the Duval and Tweedie trim-ad-fill method.

For the primary analysis, antidepressant medications were grouped into the categories of SSRI (fluvoxamine, sertraline, citalopram, fluoxetine, paroxetine, escitalopram), SNRI (venlafaxine, duloxetine, desvenlafaxine), and atypical (vilazadone, bupropion, vortioxetine, mirtazapine, trazodone). The following questions were examined in meta-analysis: (1) what is the risk for hyperhidrosis for different for SSRI, SNRI, and the atypical medications (analyzed individually) as compared to placebo (P <0.05 for SSRI, SNRI, and other medications in random-effects meta-analysis using RR as primary outcome); (2) do SSRIs and SNRIs differ in the risk of hyperhidrosis when compared to placebo (test for subgroup differences in random-effects meta-analysis with trials restricted to SSRIs and SNRIs); (3) do different pharmacological agents within a medication class (e.g., SSRI and SNRI) differ in risk of hyperhidrosis (test for subgroup difference with trials stratified by pharmacological agent and trials restricted to individual medication classes (e.g., SSRIs and SNRIs); (4) is dosing of medications (SSRIs or SNRIs) associated with the risk of hyperhidrosis (metaregression by dose with trials restricted to each medication class (e.g. SSRIs and SNRIs)); (5) does diagnostic indication for treatment affect the risk of hyperhidrosis (anxiety disorder, OCD, or major depressive disorder); and (6) are pharmacological properties of antidepressant medications (e.g., receptor affinities) associated with the risk of hyperhidrosis. All subgroup analyses were performed using a random-effects model to conduct a test for subgroup differences (between-group heterogeneity χ^2). Metaregression analysis was used to examine the effects of maximum daily dose of medication used in trials, all daily doses of medication were converted into venlafaxine (imipramine) equivalents using previously described methodology (Bollini et al., 1999). The threshold for statistical significance was P <0.05 for the primary analysis, as well as for all stratified subgroup analyses and metaregression.

Our analysis examining the association between pharmacological properties of antidepressant medication and sleep problems was based on a recent meta-analysis performed examining the association between antidepressant receptor affinities and weight gain. We obtained data on receptor affinities of different antidepressants from published data and from searching the Psychoactive Drug Screening Program Ki database (https://pdsp.med.unc.edu/kidb.php). The inhibitory constant (Ki—the concentration of a ligand in which the receptor is 50% occupied) was used as our measure of binding affinity, and consistent with previous analysis in this area, (1) only human data was used to estimate Ki, (2) all values of Ki were natural log transformed and (3) a maximum value of 10000 nM was used for low-affinity interactions. We utilized random-effects metaregression to examine the association between log Ki at receptors of interest and the relative risk of hyperhidrosis. We conducted metaregression for all receptors where data on affinity is typically available for antidepressants—specifically the serotonin transporter, norepinephrine transporter, dopamine transporter (DAT), H1-histamine receptor (H1), alpha-1 and alpha-2 adrenergic receptors, M3 muscarinic receptor, and 5-HT1A, 5-HT2A, and 5-HT2C serotonin receptors.

3 | RESULTS

3.1 | Included trials

Figure 1 depicts the selection of trials for this meta-analysis. A total of 382 references were identified in PubMed. A total of 242 trials were eligible for potential inclusion once the titles, abstracts, and full texts (where necessary) were screened. Around 160 studies did not provide adequate side-effect data on hyperhidrosis for contribution to meta-analysis outcomes. Our meta-analysis involved data from 76 trials involving 28544 subjects (Allgulander et al., 2004; Baldwin, Bobes, Stein, Scharwachter, & Faure, 1999; Baldwin, Loft, & Dragheim, 2012; Bose et al., 2008; Boulenger, Loft, & Olsen, 2014; Bradwejn et al., 2005; C. Allgulander, 2001, 2004; Chouinard, 1992; Claghorn et al., 1996; Coric et al., 2010; D. S. Baldwin, 2006; Detke et al., 2004; Dominguez, Goldstein, Jacobson, & Steinbook, 1985; Dunbar et al., 1991; Dunlop et al., 2011; Edwards & Goldie, 1993; Fabre et al., 1995; Fabre, Birkhimer, Zaborny, Wong, & Kapik, 1996; Feighner et al., 1993; Feighner, Boyer, Meredith, & Hendrickson, 1989; Gaynor et al., 2011; Gaynor et al., 2011; Gelenberg et al., 2000; Golden, Nemeroff, McSorley, Pitts, & Dube, 2002; Henigsberg, Mahableshwarkar, Jacobsen, Chen, & Thase, 2012; Hewett et al., 2009; Hewett et al., 2008; Hewett et al., 2009; Kasper et al., 2009; Katona, Hansen, & Olsen, 2012; Kiev & Feiger, 1997; Koponen et al., 2007; Lader et al., 2004; Lecrubier, Bakker, Dunbar, & Judge, 1997; Lenox-Smith & Reynolds, 2003; Lepola et al., 2003; Lepola, Bergtholdt, St. Lambert, Davy, & Ruggiero, 2004; Liebowitz et al., 2003; Liebowitz et al., 2008; Liebowitz et al., 2002; Liebowitz, Asnis, Mangano, & Tzanis, 2009; Liebowitz, Gelenberg, & Munjack, 2005; Liebowitz, Mangano, Bradwejn, & Asnis, 2005; Liebowitz, Tourian, Hwang, & Mele, 2013; Mahableshwarkar, Jacobsen, & Chen, 2013; Mahableshwarkar, Jacobsen, Chen, & Simon, 2013; Mahableshwarkar, Jacobsen, Serenko, & Chen, 2014; Mendels

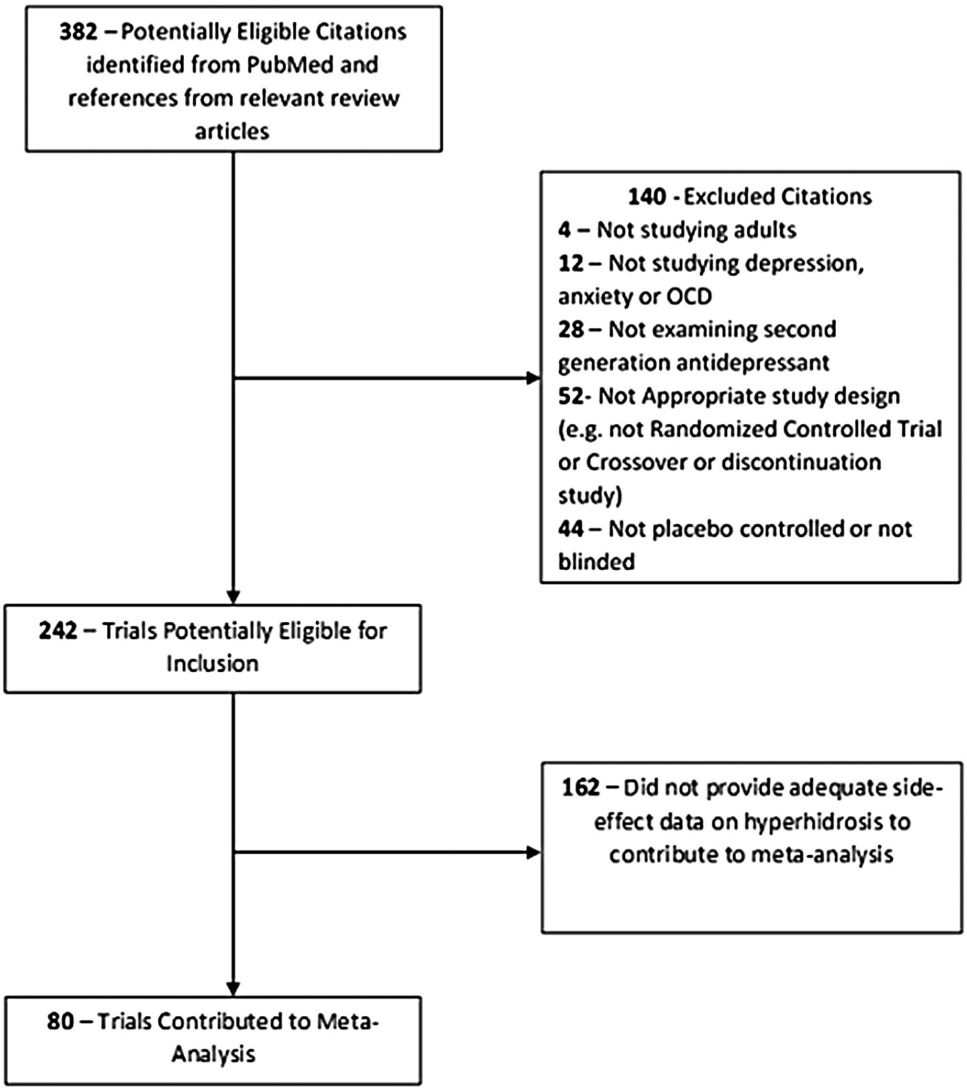

FIGURE 1 Flow diagram illustrating the selection of studies for inclusion in the meta-analysis

et al., 1999; Montgomery et al., 2001; Nicolini et al., 2008; Nierenberg et al., 2007; Nimatoudis et al., 2004; Pollack et al., 2006; Pollack et al., 1996; Reimherr et al., 1990; Reimherr, Cunningham, Batey, Johnston, & Ascher, 1998; Rickels et al., 1989; Rickels et al., 2003; Rickels, Mangano, & Khan, 2004; Rudolph & Feiger, 1999; S. Kasper, 2005; Silverstone & Ravindran, 1999; Silverstone & Salinas, 2001; Sramek et al., 1995; Stein et al., 2014; Stein et al., 2007; Stein et al., 1998; Stein, Pollack, Bystritsky, Kelsey, & Mangano, 2004; Suliman et al., 2015; Tourian, Padmanabhan, Groark, Brisard, & Farrington, 2009; Trivedi et al., 2001; Van Ameringen et al., 2001; Wu, Wang, Ball, Desaiah, & Ang, 2011; Wyeth & Pfizer, 2011; Zohar & Judge, 1996). The characteristics of included trials are depicted in Table 1.

3.2 | Hyperhidrosis

Both SSRI (RR = 2.93, 95% CI: 2.46–3.47, $z = 12.23$, $P < 0.001$) and SNRIs (RR = 3.17, 95% CI = 2.63–3.82, $z = 12.11$, $P < 0.001$) were associated with a significantly increased risk of hyperhidrosis compared to placebo. There was no evidence of a significant difference in risk of hyperhidrosis risk by antidepressant class (test for subgroup differences $\chi^2 = 0.01$, df = 1, $P = 0.91$). Figure 2 depicts the relative risk of hyperhidrosis for individual antidepressant medications as well as antidepressant medication classes (SNRI, SSRI, and atypical). All antidepressant medications were associated with a significantly increased risk of hyperhidrosis except fluvoxamine (RR = 0.56, 95% CI: 0.12–2.53, $z = -0.76$, $P = 0.45$), bupropion (RR = 1.23, 95% CI: 0.57–2.67, $z = 0.53$, $P = 0.60$), and vortioxetine (RR = 1.35 95%, CI: 0.79–2.33, $z = 1.10$, $P = 0.27$)

There was evidence of significant publication bias among SSRI [Egger's test: Intercept = 0.91 (95% CI = 0.174–1.628) $P = 0.02$], and SNRI [Egger's test: Intercept = 1.26 (95% CI = 0.57–1.95) $P < 0.001$) trials separately. Figure 3 presents a funnel plot displaying asymmetry consistent with reporting bias of hyperhidrosis side-effect data. When relative risk estimates were adjusted for funnel plot asymmetry using the Duval and Tweedie trim-ad-fill method the relative risk of hyperhidrosis all medication classes—SSRIs (RR = 2.54, 95% CI = 2.05–3.14) and SNRIs (RR = 2.56, 95% CI = 2.08–3.13)—were still associated with a significantly increased risk of hyperhidrosis.

TABLE 1 Characteristics of included trials

Study	Medication	Class	Dose (mg/d)	ED	Active: Placebo	Indication
SSRI						
Lepola et al., 2003	Citalopram	SSRI	40	133	160:154	Depression
Montgomery et al., 2001	Citalopram	SSRI	20, 40, 60	67, 133, 200	300:101	OCD
Mendels et al., 1999	Citalopram	SSRI	80	266	89:91	Depression
Suliman et al., 2015	Escitalopram	SSRI	20	133	12:17	Anxiety
Coric et al., 2010	Escitalopram	SSRI	20	133	51:101	Anxiety
Bose et al., 2008	Escitalopram	SSRI	20	133	45:52	Anxiety
Stein et al., 2007	Escitalopram	SSRI	10, 20	67, 133	227:114	OCD
Stein et al., 2007	Escitalopram	SSRI	20	133	141:131	Anxiety
Nierenberg et al., 2007	Escitalopram	SSRI	10	66.6	274:137	Depression
Baldwin, 2006	Escitalopram	SSRI	5, 10, 20	33, 67, 133	403:139	Anxiety
Kasper, 2005	Escitalopram	SSRI	20	133	181:177	Anxiety
Lader et al., 2004	Escitalopram	SSRI	5, 10, 20	33, 67, 133	504:166	Anxiety
Lepola et al., 2003	Escitalopram	SSRI	20	133	155:154	Depression
Silverstone and Salinas, 2001	Fluoxetine	SSRI	60	300	33:25	Depression
Silverstone and Salinas, 2001	Fluoxetine	SSRI	60	300	88:94	Depression
Rudolph and Feiger, 1999	Fluoxetine	SSRI	60	300	103:98	Depression
Silverstone and Ravindran, 1999	Fluoxetine	SSRI	60	300	119:118	Depression
Sramek et al., 1995	Fluoxetine	SSRI	20	100	72:72	Depression
Fabre et al., 1996	Fluvoxamine	SSRI	150	150	46:44	Depression
Feighner et al., 1989	Fluvoxamine	SSRI	300	300	31:19	Depression
Dominguez et al., 1985	Fluvoxamine	SSRI	300	300	35:31	Depression
Pollack, 2007	Paroxetine	SSRI	40	200	166:163	Anxiety
Stein et al., 2007	Paroxetine	SSRI	40	200	117:114	OCD
Baldwin, 2006	Paroxetine	SSRI	20	100	139:139	Anxiety
Liebowitz et al., 2005	Paroxetine	SSRI	50	250	142:146	Anxiety
Allgulander, 2004	Paroxetine	SSRI	50	250	144:146	Anxiety
Detke et al., 2004	Paroxetine	SSRI	20	100	86:93	Depression
Lader et al., 2004	Paroxetine	SSRI	20	67	169:166	Anxiety
Lepola et al., 2004	Paroxetine	SSRI	38	188	186:184	Anxiety
Trivedi, 2004	Paroxetine	SSRI	13, 25	63, 125	301:146	Depression
Rickels et al., 2003	Paroxetine	SSRI	20, 40	100, 200	386:180	Anxiety
Golden et al., 2002	Paroxetine	SSRI	50, 63	250, 313	429:211	Depression
Liebowitz et al., 2002	Paroxetine	SSRI	20, 40, 60	100, 200, 300	289:95	Anxiety
Baldwin et al., 1999	Paroxetine	SSRI	50	250	139:151	Anxiety
Lecrubier et al., 1997	Paroxetine	SSRI	20	100	123:123	Anxiety
Stein et al., 1998	Paroxetine	SSRI	50	250	94:93	Anxiety
Claghorn et al., 1996	Paroxetine	SSRI	150	150	32:26	Depression
Zohar and Judge, 1996	Paroxetine	SSRI	60	300	201:99	OCD
Edwards and Goldie, 1993	Paroxetine	SSRI	30	150	21:20	Depression
Feighner et al., 1993	Paroxetine	SSRI	50	250	240:240	Depression
Kiev, 1992	Paroxetine	SSRI	50	250	38:40	Depression
Dunbar et al., 1991	Paroxetine	SSRI	50	250	240:240	Depression
Rickels et al., 1989	Paroxetine	SSRI	50	250	55:56	Depression
Allgulander, 2004	Sertraline	SSRI	150	180	184:189	Anxiety
Liebowitz et al., 2003	Sertraline	SSRI	200	240	205:196	Anxiety
Van Amerigen, 2001	Sertraline	SSRI	200	240	135:169	Anxiety

(Continues)

TABLE 1 (Continued)

Study	Medication	Class	Dose (mg/d)	ED	Active: Placebo	Indication
Fabre et al., 1995	Sertraline	SSRI	200	240	91:91	Depression
Chouinard, 1992	Sertraline	SSRI	200	240	260:260	OCD
Reimherr et al., 1990	Sertraline	SSRI	200	240	149:150	Depression
SNRI						
Liebowitz et al., 2013	Desvenlafaxine	SNRI	50, 100	100, 200	450:223	Depression
Dunlop et al., 2011	Desvenlafaxine	SNRI	50	100	285:142	Depression
Wyeth/Pfizer, 2011	Desvenlafaxine	SNRI	50	100	285:142	Depression
Tourian et al., 2009	Desvenlafaxine	SNRI	50, 100	100, 200	298:161	Depression
Liebowitz et al., 2008	Desvenlafaxine	SNRI	50, 100	100, 200	299:152	Depression
Boulenger et al., 2014	Duloxetine	SNRI	60	150	147:158	Depression
Mahableshwarkar et al., 2014	Duloxetine	SNRI	60	150	154:155	Anxiety
Mahableshwarkar et al., 2013	Duloxetine	SNRI	60	150	150:151	Depression
Gaynor et al., 2011a	Duloxetine	SNRI	60	150	261:266	Depression
Gaynor et al., 2011b	Duloxetine	SNRI	60	150	262:266	Depression
Wu et al., 2011	Duloxetine	SNRI	120	300	108:102	Anxiety
Nicolini, 2009	Duloxetine	SNRI	20, 120	50, 300	242:170	Anxiety
Tourian et al., 2009	Duloxetine	SNRI	60	150	157:161	Depression
Koponen et al., 2007	Duloxetine	SNRI	60, 120	150, 300	338:175	Anxiety
Nierenberg et al., 2007	Duloxetine	SNRI	60	150	273:137	Depression
Detke et al., 2004	Duloxetine	SNRI	120	300	120:93	Depression
Kasper et al., 2009	Venlafaxine	SNRI	225	225	125:128	Anxiety
Nicolini, 2009	Venlafaxine	SNRI	225	225	169:170	Anxiety
Liebowitz et al., 2009	Venlafaxine	SNRI	225	225	164:159	Anxiety
Bose et al., 2008	Venlafaxine	SNRI	225	225	37:52	Anxiety
Pollack, 2007	Venlafaxine	SNRI	150	150	334:163	Anxiety
Bradwejn et al., 2005	Venlafaxine	SNRI	225	225	177:178	Anxiety
Liebowitz et al., 2005a	Venlafaxine	SNRI	225	225	139:140	Anxiety
Liebowitz et al., 2005b	Venlafaxine	SNRI	225	225	141:146	Anxiety
Nimatoudis et al., 2004	Venlafaxine	SNRI	150	150	24:22	Anxiety
Rickels et al., 2004	Venlafaxine	SNRI	225	225	137:135	Anxiety
Stein et al., 2004	Venlafaxine	SNRI	75, 225	75, 225	132:73	Anxiety
Lenox-Smith et al., 2003	Venlafaxine	SNRI	150	150	122:122	Anxiety
Allgulander, 2001	Venlafaxine	SNRI	38, 75, 150	38, 75, 150	411:130	Anxiety
Silverstone and Salinas, 2001a	Venlafaxine	SNRI	225	225	34:35	Depression
Silverstone, 2001b	Venlafaxine	SNRI	225	225	94:94	Depression
Gelenberg et al., 2000	Venlafaxine	SNRI	225	225	124:127	Anxiety
Rudolph and Feiger, 1999	Venlafaxine	SNRI	225	225	100:98	Depression
Silverstone and Ravindran, 1999	Venlafaxine	SNRI	225	225	122:118	Depression
Pollack et al., 1996	Venlafaxine	SNRI	225	225	13:13	Anxiety
Atypical						
Hewett, 2010a	Bupropion	Atypical	300	129	203:187	Depression
Hewett, 2010b	Bupropion	Atypical	300	129	211:207	Depression
Hewett et al., 2009	Bupropion	Atypical	150	65	187:197	Depression
Reimherr et al., 1998	Bupropion	Atypical	150, 300	65, 129	236:117	Depression
Boulenger et al., 2014	Vortioxetine	Atypical	15, 20		302:158	Depression
Mahableshwarkar, 2014	Vortioxetine	Atypical	3, 10		303:153	Anxiety
Mahableshwarkar, 2014	Vortioxetine	Atypical	3, 5, 10		467:155	Depression

(Continues)

TABLE 1 (Continued)

Study	Medication	Class	Dose (mg/d)	ED	Active: Placebo	Indication
Mahableshwarkar et al., 2013	Vortioxetine	Atypical	3, 5		302:151	Anxiety
Baldwin et al., 2012	Vortioxetine	Atypical	3, 5, 10		463:148	Depression
Henigsberg et al., 2012	Vortioxetine	Atypical	1, 5, 10		420:140	Depression
Katona et al., 2012	Vortioxetine	Atypical	5		150:157	Depression

This table reports the characteristics of trials included in this meta-analysis. References to be included as trials listed as supplementary references 1–80. ED = imipramine (or fluvoxamine) equivalent dose; SSRI = selective serotonin reuptake inhibitor, and SNRI = serotonin norepinephrine reuptake inhibitor.

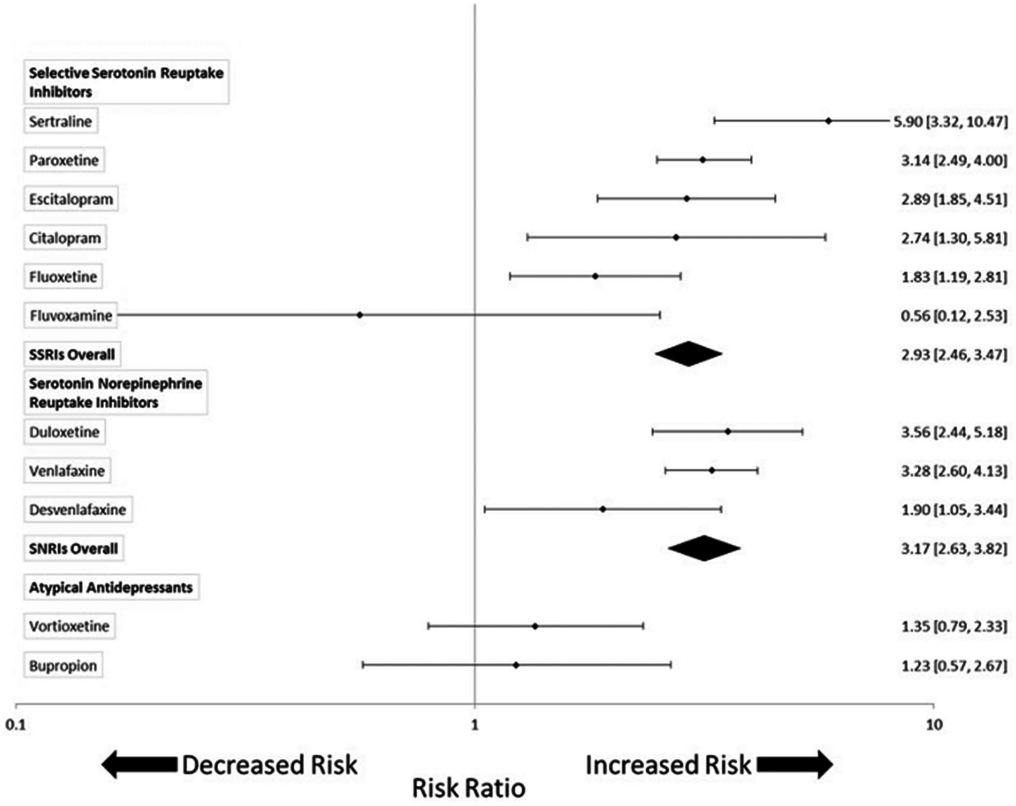

FIGURE 2 Forest plot demonstrating the relative risk of hyperhidrosis among SSRI, SNRI and atypical agents. A forest plot of the relative risk of hyperhidrosis in each studied medication, within studied class (SSRI, SNRI, and Atypical), plotted logarithmically. Medication specific relative risks (including 95% CI) are donated by black lines. Class specific relative risk (including 95% CI) is represented by a black diamond

3.2.1 | Selective serotonin reuptake inhibitors

Selective serotonin reuptake inhibitors were associated with a significantly increased risk of hyperhidrosis when compared to placebo (RR = 2.93, 95% CI: 2.46–3.47, z = 12.23, P < 0.001). There was a modest amount of heterogeneity between trials (test for heterogeneity $\chi^2 = 57.9$, df = 47, $P = 0.13$, $I^2 = 18.8\%$) with evidence of publication bias as discussed earlier.

Stratified subgroup analysis demonstrated significant differences in relative risk of hyperhidrosis between SSRI medications as compared to placebo (test for subgroup differences χ^2 = 15.3, df = 5, P = 0.009). Fluvoxamine was not associated with an increased risk of hyperhidrosis when compared to placebo (RR = 0.56, 95% CI: 0.12–2.53, z = −0.76, P = 0.45). All other SSRI medications were associated with a significantly increased risk of hyperhidrosis compared to placebo (sertraline (RR = 5.90, 95% CI: 3.32–10.47, $z = 6.06$, $P < 0.001$); paroxetine (RR = 3.14, 95% CI: 2.49–4.00, z = 9.56, P < 0.001); escitalopram (RR = 2.89, 95% CI = 1.85–4.51, z = 4.68, P < 0.001); citalopram (RR = 2.74, 95% CI = 1.30–5.81, z = 2.64, P = 0.008); fluoxetine (RR = 1.83, 95% CI = 1.19–2.81, z = 2.74, P = 0.006)). Risk of hyperhidrosis with SSRI medications did not differ based on diagnostic indication (test for subgroup differences $\chi^2 = 2.9$, df = 2, $P = 0.24$). Metaregression demonstrated no significant association between dosing of SSRI medications and the risk of hyperhidrosis compared to placebo (Parameter Estimate (PE) = −0.0013, 95% CI: −0.0032–0.007, $z = -1.3$, $P = 0.19$).

3.2.2 | Serotonin–norepinephrine reuptake inhibitors

Serotonin–norepinephrine reuptake inhibitors were associated with a significantly increased risk of hyperhidrosis when compared to placebo (RR = 3.17, 95% CI = 2.63–3.82, z = 12.11, P < 0.001). There was no evidence of heterogeneity between trials (test for heterogeneity

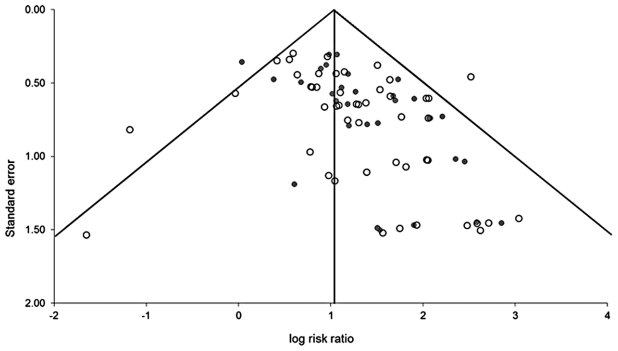

FIGURE 3 Funnel plot examining publication bias for risk of hyperhidrosis. Figure examines publication bias using a funnel plot which depicts log transformed risk ratio of hyperhidrosis compared to placebo (x axis) and standard error of included trials (y axis). There was evidence of significant publication bias among SSRI [Egger's test: Intercept = 0.91 (95% CI = 0.174–1.628) $P = 0.02$), and SNRI [Egger's test: Intercept = 1.26 (95% CI = 0.57–1.95) $P < 0.001$) trials separately, as well as asymmetry of the funnel plot

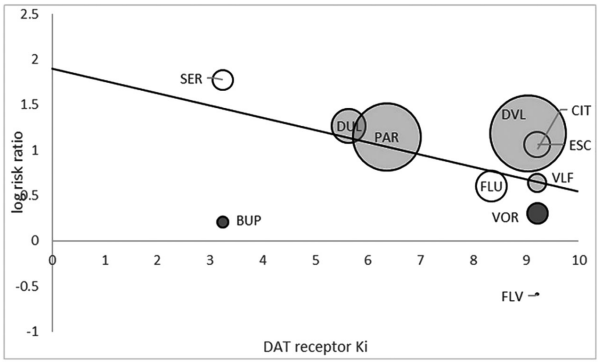

FIGURE 4 Association between dopamine transporter (DAT) affinity and risk of hyperhidrosis. Figure demonstrates a scatterplot showing the correlation between receptor affinity (log transformed ki value–*x* axis) for DAT receptors and log transformed relative risk of hyperhidrosis (y axis) for each medication. The circle size is proportional to weight of each medication within the analyses. CIT–citalopram; ESC–escitalopram; FLU–fluoxetine; FLV–fluvoxamine; PAR–paroxetine; SER–sertraline; DVL–desvenlafaxine; DUL–duloxetine; VLF–venlafaxine; BUP–bupropion; VOR–vortioxetine

$\chi^2 = 30.9$, df = 33, $P = 0.57$, $I^2 = 0\%$) but there was evidence of publication bias within SNRI trials.

There was no significant difference in risk of hyperhidrosis compared to placebo between different SNRI medications (test for subgroup differences $\chi^2 = 3.29$, df = 2, $P = 0.19$). The relative risk for hyperhidrosis of each SNRI medication as compared to placebo was duloxetine (RR = 3.56, 95% CI: 2.44–5.18, $z = 6.60$, $P < 0.001$), venlafaxine (RR = 3.28, 95% CI: 2.60–4.13, z = 10.10, $P < 0.001$), and desvenlafaxine (RR = 1.90, 95% CI: 1.05–3.44, $z = 2.11$, $P = 0.04$). Risk of hyperhidrosis with SNRI medications did not differ based on diagnostic indication (test for subgroup differences $\chi^2 = 2.00$, df = 1, $P = 0.16$). Metaregression demonstrated no significant association between dosing of SNRI medications and the risk of hyperhidrosis compared to placebo (PE = 0.0019, 95% CI: −0.0009–0.0046, $z = 1.3$, $P = 0.19$).

3.2.3 | Atypicals

Neither bupropion (RR = 1.23, 95% CI: 0.57–2.67, $z = 0.53$, $P = 0.60$) nor vortioxetine (RR = 1.35, 95% CI: 0.79–2.33, $z = 1.10$, $P = 0.27$) were associated with an increased risk of hyperhidrosis compared to placebo. Randomized trials of other atypical antidepressant medications (mirtazapine, vilazodone, and trazodone) did not report sufficient data on hyperhidrosis as a side effect.

3.2.4 | Receptor analysis of SRIs

Increased DAT affinity of antidepressant medications was associated with relative risk of hyperhidrosis compared to placebo (PE = −0.09, 95% CI: −016–0.02, $z = -2.61$, $P = 0.009$). Figure 4 depicts a scatterplot of the significant association between DAT receptor affinity of SRI medications and the relative risk of hyperhidrosis compared to placebo. This association remained significant, and even increased, after controlling for dosing of SRI medications (PE = −0.21, 95% CI: −0.32–0.10, $z = -3.73$, $P < 0.001$). Serotonin–norepinephrine reuptake inhibitors medication affinity for other studied receptors was not significantly associated with the relative risk of hyperhidrosis. [(serotonin transporter (PE = −0.01, 95% CI: −0.08–0.02, $z = -0.45$, $P = 0.65$), norepinephrine transporter (PE = −0.02, 95% CI: −0.09–0.04, $z = -0.64$, $P = 0.52$), H1 (PE = 0.09, 95% CI: −0.06–0.25, $z = 1.16$, $P = 0.25$), M3 muscarinic receptor (PE = −0.001, 95% CI: 0.036–0.07, $z = -0.02$, $P = 0.98$), Alpha1 (PE = −0.08, 95% CI: −0.20–0.05, $z = -1.24$, $P = 0.22$), Alpha2 (PE = −0.12, 95% CI: −0.56–0.33, $z = -0.51$, $P = 0.61$), 5-HT1A (PE = −0.20, 95% CI: −0.80–0.39, $z = -0.67$, $P = 0.50$), 5-HT2A (PE = 0.01, 95% CI: −0.08–0.10, $z = 0.22$, $P = 0.83$), and 5-HT2C (PE = 0.05, 95% CI: −0.06–0.17, $z = 0.88$, $P = 0.38$).

4 | DISCUSSION

Meta-analysis demonstrated that most antidepressant medications were associated with an increased risk of hyperhidrosis compared to placebo. Both SSRIs and SNRIs demonstrated similar risks of hyperhidrosis compared to placebo. Doses of SSRI and SNRIs and diagnostic indication for treatment were not associated with differences in risk of hyperhidrosis. There were also significant differences between individual SSRI medications and risk of hyperhidrosis with fluvoxamine appearing to carry a lower risk, and sertraline and paroxetine carrying a higher risk than other SSRIs. Metaregression suggested that risk of hyperhidrosis for SRI medications was associated with increased affinity for the DAT.

Hyperhidrosis is a side effect of SSRI and SNRI medications as well as several other antidepressant medications (e.g., tricyclic antidepressants) not included in this review (Cheshire & Fealey, 2008; Marcy & Britton, 2005; Preskorn, 1995). Although the exact mechanism of ADIES is unclear, it is hypothesized to be related to serotonin's effects on thermoregulation in the hypothalamus. Increases in extracellular serotonin in the hypothalamus of rats has been demonstrated to induce hyperthermia (Lin, Tsay, Su, & Chueh, 1998). The clearest

demonstration of a similar effect in humans are the symptoms of serotonin syndrome (Donoghue, Tylee, & Wildgust, 1996). Serotonin syndrome is a group of symptoms that may occur following use of certain serotonergic medications (including antidepressants) or drugs (Donoghue et al., 1996). Serotonin syndrome and its spectrum of symptoms are a product of hyperstimulation of both central and peripheral serotonin receptors, particularly 5-HT2A (Boyer & Shannon 2005). Diaphoresis and hyperthermia are principle symptoms of serotonin syndrome along with hyperreflexia, diarrhea, agitation, seizures, and pupillary dilation (Boyer & Shannon 2005).

Our metaregression suggests that hyperhidrosis risk among antidepressant medications was associated with decreased DAT affinity and inhibition. Agents with low-DAT affinity, such as fluvoxamine, (es)citalopram, and venlafaxine were among the least likely to induce hyperhidrosis. Dopamine, like serotonin, is critical for thermoregulation in the hypothalamus (Lee, Mora, & Myers, 1985). Similar to serotonin syndrome, diaphoresis and hyperthermia are association with conditions caused by excessive dopamine, such as neuroleptic malignant syndrome (Gurrera, 1999). Hyperhidrosis is also a common symptom of another disorder associated with dopamine dysregulation–Parkinson's disease (Swinn et al., 2003). In Parkinson's disease (PD), dopamine nerve terminals and consequently DAT are reduced by 30–50% in early PD and to a greater extent in more severe PD (Marek et al., 1996; Rinne et al., 1999). Excessive sweating may be experienced by as many as two-thirds of individuals with PD and is more common with progression of the illness (Swinn et al., 2003). Aripiprazole, a partial agonist of the D2 receptor, has also been suggested as a possible treatment for ADIES in case reports (Lu, Cullen, Eide, Williams, & Apfeldorf, 2008).

Although the mechanism of ADIES is unclear and likely multifactorial, several treatments exist for patients to reduce excessive sweating. Terazosin is an alpha-1 agonist that has been demonstrated to significantly reduce ADIES in randomized and placebo-controlled trials (Ghaleiha, Shahidi, Afzali, & Matinnia, 2013). Terazosin is hypothesized to reduce ADIES by decreasing sympathetic tone at the level of the hypothalamus (Mago et al., 2013). Other medications, such as benzotropine, glycopyrrolate, cyproheptadine, and aripiprazole have been reported in case reports or case series to improve ADIES. Benzotropine and glycopyrrolate are anticholinergic agents that have been used effectively to control ADIES (Kolli & Ramaswamy, 2013; Mago, 2013). Specifically, these agents are believed to reduce excessive sweating by the direct inhibition of sweat glands that are controlled peripherally by cholinergic nerve fibers (Marcy & Britton, 2005; Marek et al. 1996; Swinn et al., 2003). Cyproheptadine, which is an antagonist of acetylcholine, serotonin and histamine has also been utilized to control ADIES (Ashton & Weinstein, 2002). Cyproheptidine, via 5-HT2A antagonism, may effect serotonergic tone in the hypothalamus in addition to having peripheral anticholinergic effects that reduce excessive sweating.

It is important to acknowledge the limitations of this meta-analysis. There is inconsistency in the reporting of hyperhidrosis as a side effect in antidepressants, and the reporting is done in a nonrandom, biased manner, favoring reporting of side effects when significant differences from placebo are observed. As such we demonstrated that there was a reporting bias on hyperhidrosis as a side effect in antidepressant trials across medication classes. Additionally, patients and subjects often do not report hyperhidrosis as a side effect. This may be due to embarrassment, or not making the connection between increased sweating and the medication. Thus either clinical trials employing rating scales that measure hyperhidrosis (such as the Hyperhidrosis Disease Severity Scale) or use of physiologic measures of sweating in response to challenge or treatment with antidepressant medications would be more sensitive to detecting differences between medications (Kowalski, Eadie, Dagget, & Lai,). Furthermore, methods of rating side effects across trials may differ and thus differences in the risk of hyperhidrosis between agents may not only represent actual differences in the risk between agents but also possibly systematic differences in assessment method. While the gold standard method for determining differences in side effects between medications would be head-to-head trials rather than indirect comparison using a placebo, such trials are not widely available in the literature. Likewise, the association between antidepressant dosing and the risk of hyperhidrosis would also ideally be examined using fixed-dosage trials. Metaregression examining the influence of dosage across antidepressants using imipramine equivalents could lead to false-negative findings if the dose–response relationship is different among antidepressants within the same class or if the risk of hyperhidrosis between agents is not related to imipramine equivalency.

This meta-analysis demonstrated (1) that SSRIs and SNRIs are both associated with similarly increased rates of hyperhidrosis (2) there is a significant difference in the relative risk for hyperhidrosis between medications within the SSRI class, while we found no demonstrated significant difference between medications within the SNRI class; (4) there was no evidence of a relationship of dosing and risk for hyperhidrosis with both SSRIs and SNRIs; (5) there was no difference in relative risk of hyperhidrosis with SSRIs or SNRIs based on diagnostic indication; and (6) increased affinity for the DAT receptor was associated with an increased relative risk of hyperhidrosis among all SSRI and SNRI medications.

DISCLOSURES

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